

Lesson 15 — Design Challenge — Press-to-Go Gadget

Matches LEGO Teacher Portal lesson: "Animal Band".

Objective (I can...): I can design a gadget that uses events to make the motor go, and I can program a motor with my chosen events. **CS concept:** Programming a motor with events (applying events to a purposeful build) · **Builds on:** Lessons 11-14. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min, read aloud): A smart doorbell reacts to events — motion AND a face it knows. *Ask the group:* "For YOUR gadget, should it need both triggers, or just one?"

Materials: LEGO CS & AI kit + device per pair, plan mats, color cards/trigger objects. **Model already built** (Events model from Lesson 11) — the 25 min is plan + code + play, small changes only.

60-minute flow - Warm-up (10): Rapid review — pairs name every event they used (press, tap, color, sensor) and act one out. *(Theme: Gadget fair.)* - **Concept + demo (8):** Challenge: build a "press-to-go gadget" that makes the motor do something fun when two different events happen. Coach says a plan for one event, then steps back. *(Try "the silly robot.")* - **Build/Code (25):** Pairs plan and code two events that each drive the motor (press → spin a spinner; color seen → open a gate) on the built model. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a third event or make one motor response a two-step move. - **Share + tidy-up (10): Gadget fair** — before a pair sets off their gadget, the audience guesses which events make the motor go, then the pair triggers them live to check. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator works the triggers and points to the plan; swap at the wiggle break. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a one-event target with a partial plan. *Extension / fast-finisher* — require a color event AND a press event. *Reteach* — reuse a Lesson 13-14 program as a starting template.

Exit check: Pair shows a working gadget where two+ events each make the motor go, with a matching plan. **Progress layer:** Send home the **Unit 3 parent note** ("your child can make a model sense and react to events") with the **build photo**. **At-home (optional):** Invent a gadget for something at home; name the events that make it go. ~5 min.
